Ex.no: 5 **Date:** _____

CREATE A CD PIPELINE IN JENKINS

Aim:

To create a Jenkins pipeline that automatically checks out, builds, and tests an application, and sends an email notification with the result of each build.

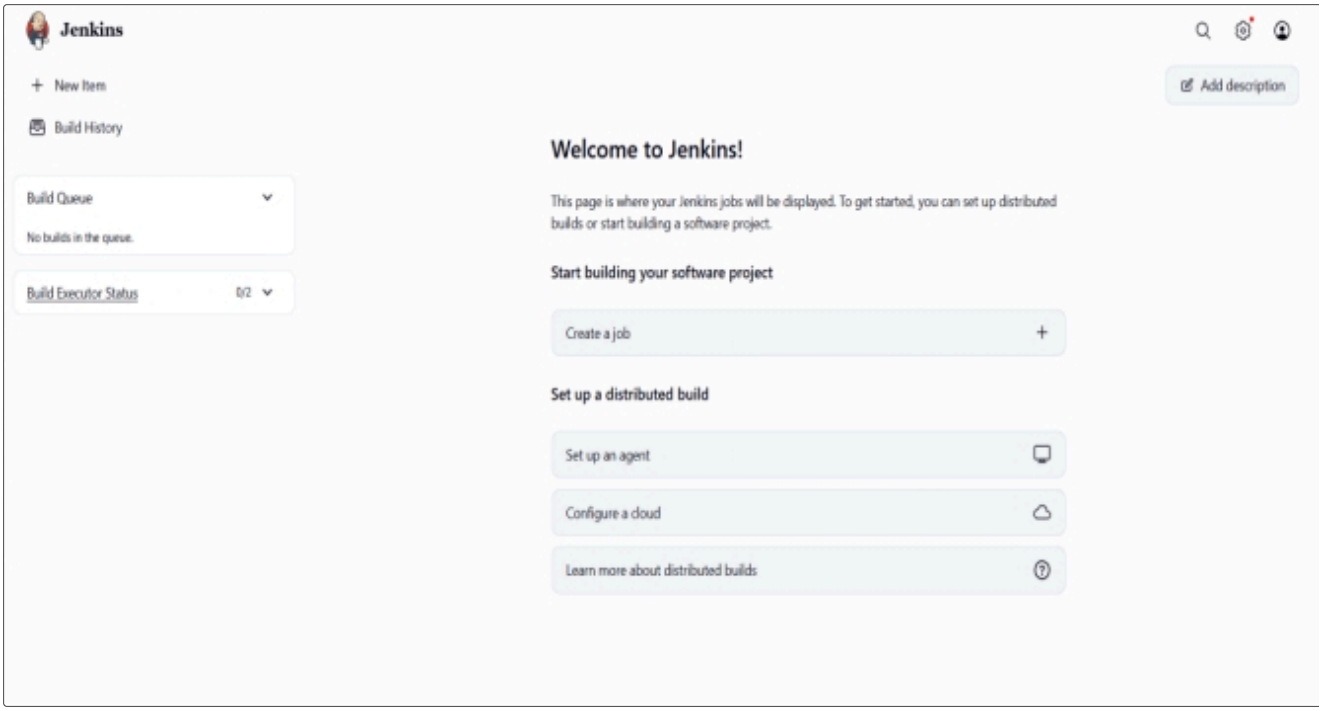
Procedure:

1. Run `sudo systemctl start jenkins` in Ubuntu, then open `http://172.19.68.232:8080` in your browser.

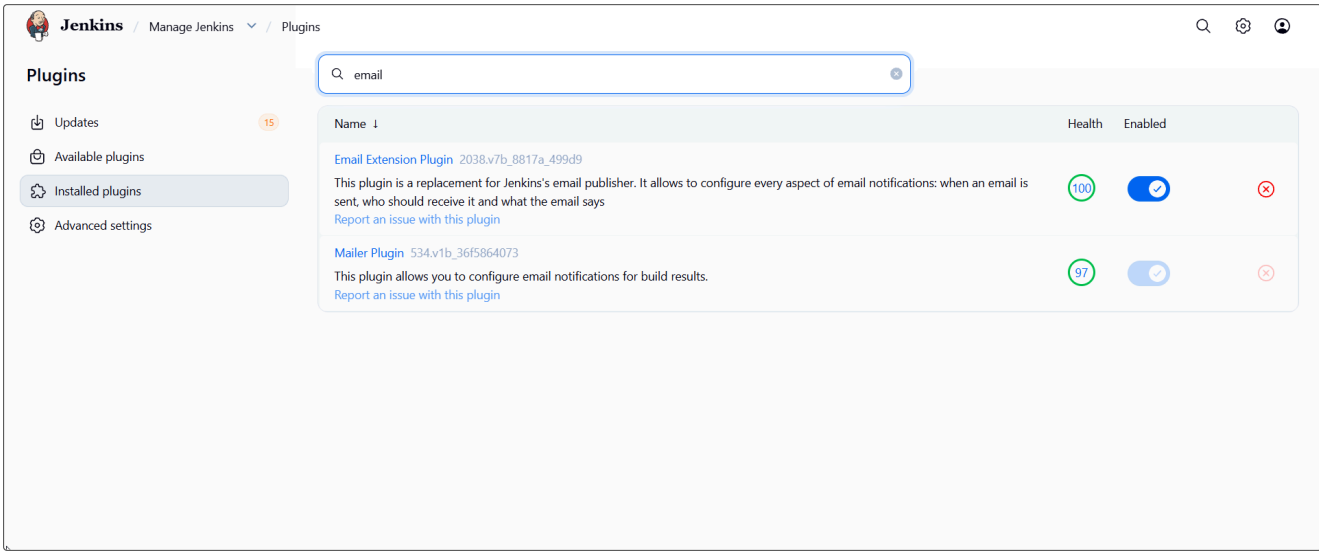
```
student@ubuntu:~$ sudo systemctl start jenkins
student@ubuntu:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-10-22 18:04:34 UTC; 2min 13s ago
   Main PID: 233 (java)
      Tasks: 54 (limit: 9303)
     Memory: 616.6M (peak: 625.5M)
        CPU: 34.892s
     CGroup: /system.slice/jenkins.service
             └─233 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins>

Oct 22 18:04:34 LAPTOP-7AE9QLDF jenkins[233]: 2026-10-22 18:04:34.148+0000 [id=73]        INFO        jenkins.InitReact>
Oct 22 18:04:34 LAPTOP-7AE9QLDF jenkins[233]: 2026-10-22 18:04:34.159+0000 [id=57]        INFO        jenkins.InitReact>
Oct 22 18:04:34 LAPTOP-7AE9QLDF jenkins[233]: 2026-10-22 18:04:34.178+0000 [id=88]        INFO        hudson.util.Retri>
Oct 22 18:04:34 LAPTOP-7AE9QLDF jenkins[233]: 2026-10-22 18:04:34.231+0000 [id=60]        INFO        jenkins.InitReact>
Oct 22 18:04:34 LAPTOP-7AE9QLDF jenkins[233]: 2026-10-22 18:04:34.280+0000 [id=40]        INFO        hudson.lifecycle.>
Oct 22 18:04:34 LAPTOP-7AE9QLDF systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Oct 22 18:04:42 LAPTOP-7AE9QLDF jenkins[233]: 2026-10-22 18:04:42.240+0000 [id=88]        INFO        h.m.DownloadServi>
Oct 22 18:04:43 LAPTOP-7AE9QLDF jenkins[233]: 2026-10-22 18:04:43.930+0000 [id=88]        INFO        h.m.DownloadServi>
Oct 22 18:04:45 LAPTOP-7AE9QLDF jenkins[233]: 2026-10-22 18:04:45.706+0000 [id=88]        INFO        h.m.DownloadServi>
Oct 22 18:04:45 LAPTOP-7AE9QLDF jenkins[233]: 2026-10-22 18:04:45.707+0000 [id=88]        INFO        hudson.util.Retri>
lines 1-20/20 (END)...skipping...
```

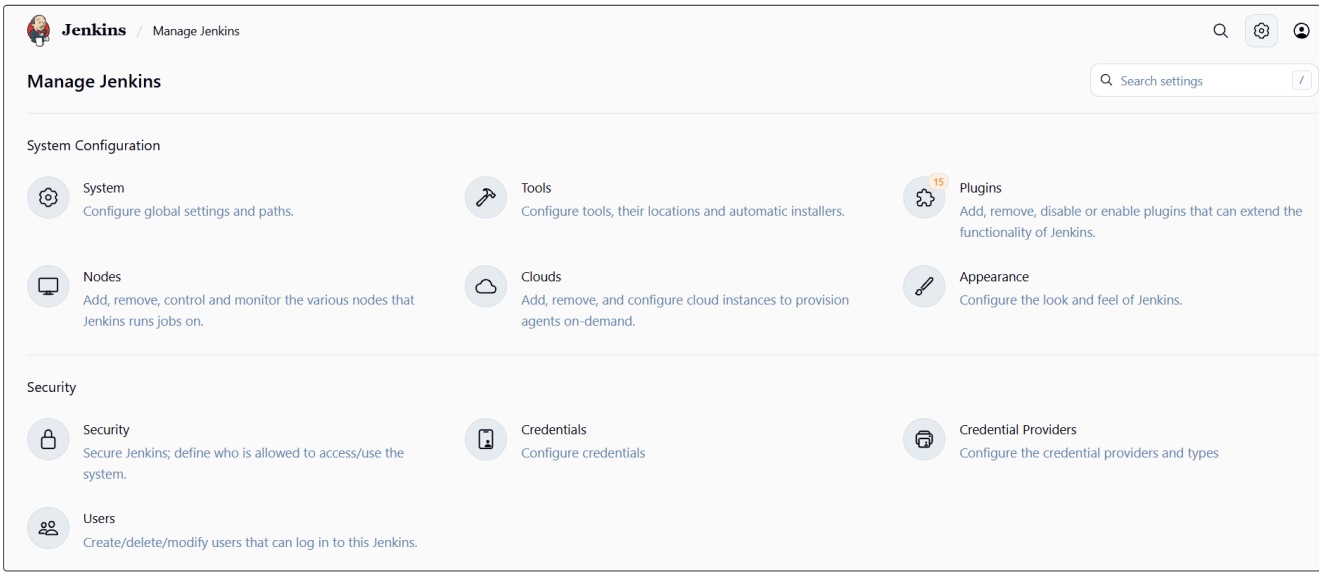
2. Open `http://172.19.68.232:8080` in a browser and log in with administrator credentials to access the Jenkins dashboard.



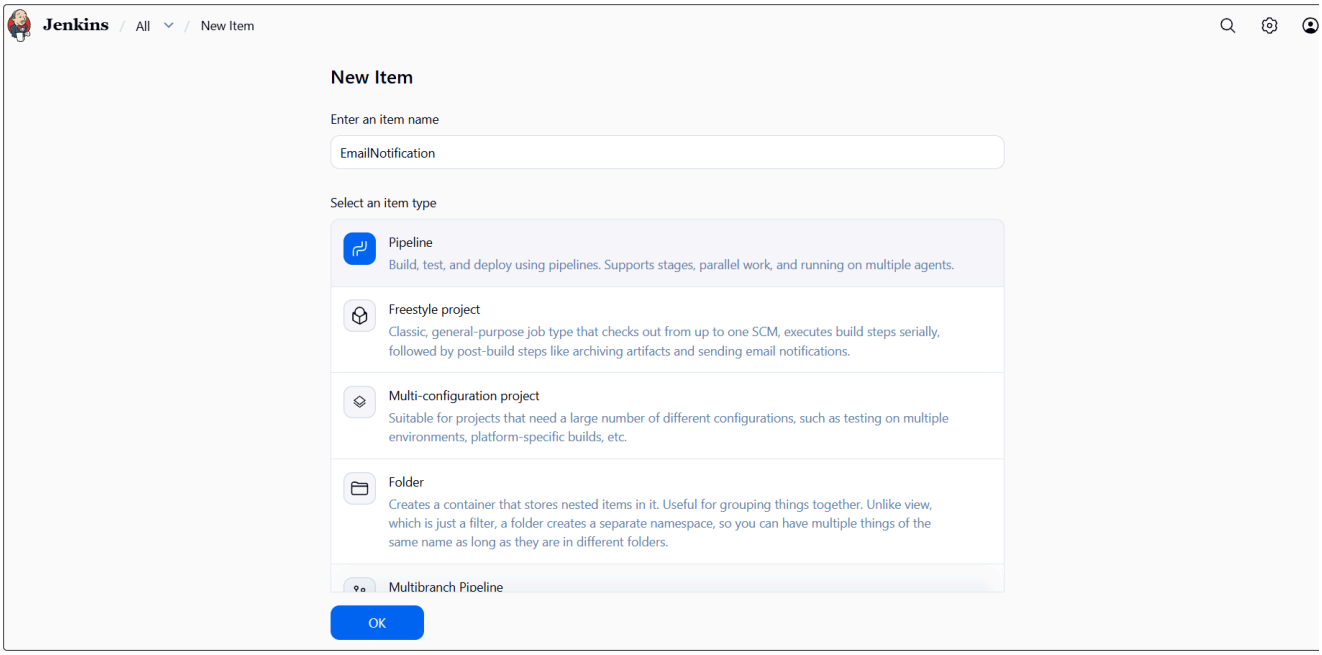
3. Go to **Manage Jenkins → Manage Plugins**, search for **Email Extension Plugin** in the Available tab, and install it if not already installed.



4. Go to **Manage Jenkins → System → E-mail Notification**, enter `localhost` as the SMTP server, click Advanced, untick *Use SMTP Authentication* and *Use SSL*, set the SMTP port to `1025`, then click Apply and Save.

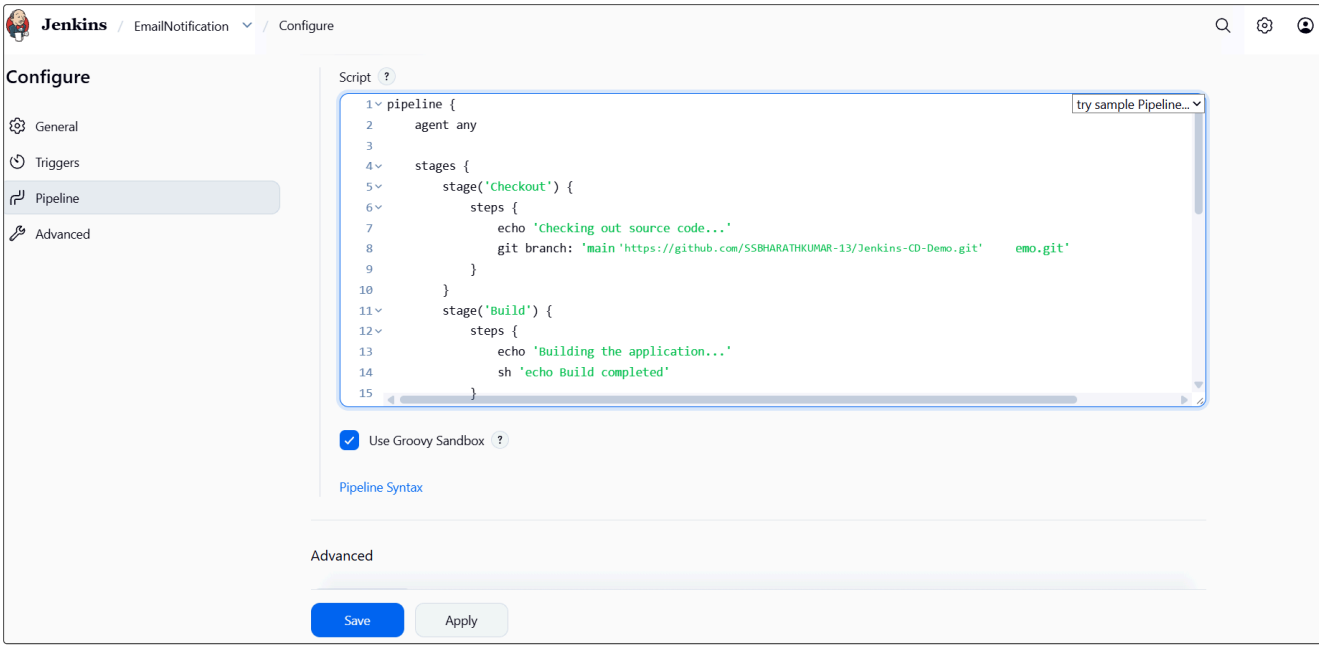


5. **Create Jenkins Pipeline Job:** Now go to Jenkins homepage and create a new job. Name the job with whatever name that you want (e.g., `EmailNotification`) and select **Pipeline**. Click on OK.

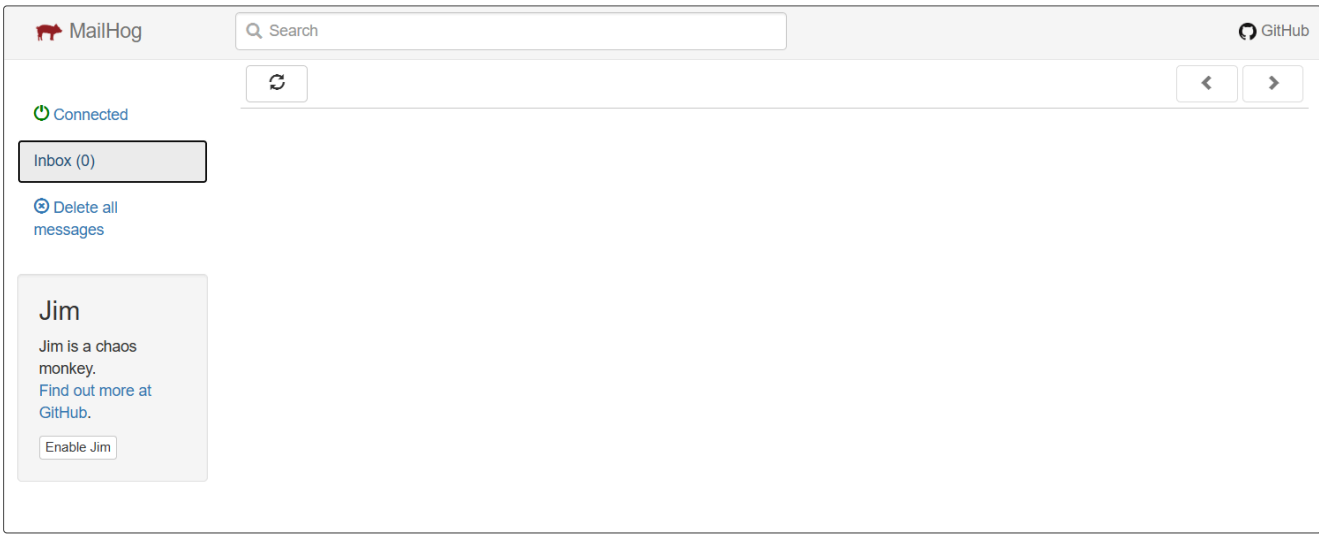


Now in the pipeline section type the following code:

```
pipeline {
    agent any
    stages {
        stage('Checkout') {
            steps {
                echo 'Checking out source code...'
                git branch: 'main', url: 'https://github.com/SSBHARATHKUMAR-13/Jenkins-CD-Demo.git'
            }
        }
        stage('Build') {
            steps {
                echo 'Building the application...'
                sh 'echo Build completed'
            }
        }
        stage('Test') {
            steps {
                echo 'Running tests...'
                sh 'echo Tests passed'
            }
        }
    }
    post {
        success {
            mail to: 'ssbharathkumar03@gmail.com',
                 subject: "Build #${env.BUILD_NUMBER} SUCCESS - ${env.JOB_NAME}",
                 body: "The pipeline ${env.JOB_NAME} (build #${env.BUILD_NUMBER}) completed successfully. Details: ${env.BUILD
        }
        failure {
            mail to: 'ssbharathkumar03@gmail.com',
                 subject: "Build #${env.BUILD_NUMBER} FAILED - ${env.JOB_NAME}",
                 body: "The pipeline ${env.JOB_NAME} (build #${env.BUILD_NUMBER}) failed. Check the console: ${env.BUILD_URL}c
        }
    }
}
```



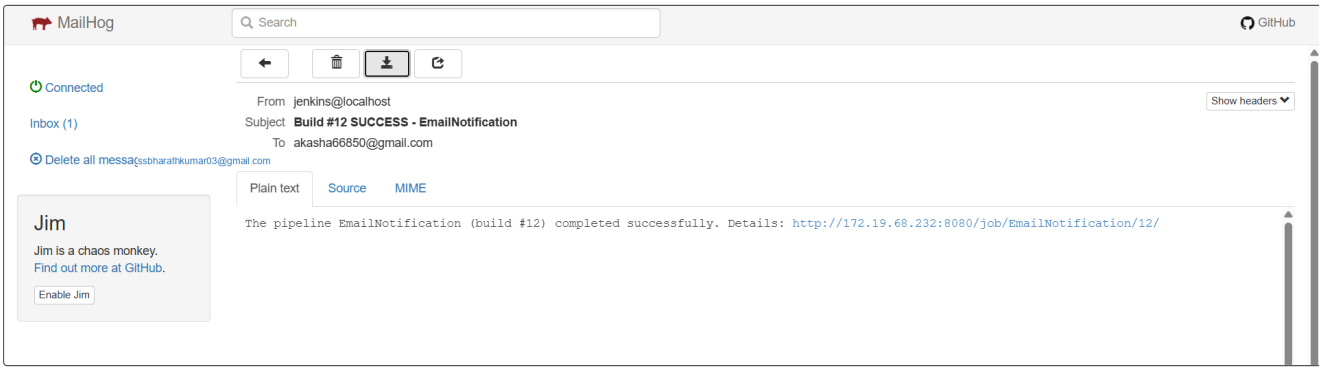
6. Run `sudo docker run -d -p 1025:1025 -p 8025:8025 mailhog/mailhog` in a terminal to start the local mail server, then open `http://localhost:8025` in a browser to see its empty inbox.



7. Execute the Jenkins pipeline and verify that all stages—**Checkout, Build, Test, and Post Actions**—complete successfully, with the email notification step executed.



8. After the Jenkins pipeline completes successfully, open the MailHog inbox at `http://localhost:8025`. The email notification with the subject “Build #12 SUCCESS - EmailNotification” confirms that the Build, Test, and Deploy stages have completed successfully.



Department of Computer Science and Engineering Rubrics for Lab Experiment		
Evaluation Parameter	**Max Marks**	**Marks Awarded**
Record	**20**	
Conduct of Experiment	**30**	
Result & Output	**20**	
Viva	**30**	
Total Marks	**100**	

Result:

The CD pipeline was executed successfully in Jenkins, and an email notification confirmed successful deployment.